ROLLOMATIC®
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Le Landeron, 19th March 2026

MASTER INTERNSHIP CERTIFICATE

This is to certify that **Ahmed Aziz Ben Haj Hmida**, born on **August 8th, 2000**, successfully completed an industrial Master Internship and a Master Thesis from **February 17th, 2025** to **March 11th, 2026** in our Innovation Cell at **EPFL**.

During this period, Aziz has demonstrated exceptional technical and research skills by developing an advanced **Curve Search Engine** that streamlines the design process for cutting tools and optimizes tool creation.

Aziz's project involved the **end-to-end** implementation of a solution for identifying geometrically and functionally similar laser-machining tools, helping reduce redundant designs and enhance engineering workflows. His work included constructing a **dataset with expert annotation**, extracting **curve-level features**, and applying **deep learning techniques**, such as rank-aware metric learning, to improve similarity evaluation.

Aziz focused on selecting the most effective approach for curve representation and similarity learning. He analyzed geometric descriptors, deep models, and implemented ranking metrics for comparing tool curves. His research establishes a solid foundation for future industrial retrieval systems, showcasing his ability to translate theoretical concepts into practical solutions, and earning positive feedback from engineers for his rigor, independence, and meaningful contributions.

His dedication and determination added significant value to our company. We were highly satisfied with his flawless execution of assigned tasks.

Upon completing his internship, Ahmed Aziz Ben Haj Hmida left the company with no outstanding commitments, apart from his obligation to maintain strict confidentiality regarding Rollomatic products and internal data.

We extend our sincere gratitude to Aziz for his excellent work and collaboration during his time with us and wish him great success in his future endeavors.

ROLLOMATIC SA

Roland Sieber
HR Manager

Emna Maghrebi
Data & AI Specialist